

Ansford Manufacturing Group

Quality Department - Line 2 | Measurement System Analysis

Gauge R&R Study - Summary Report

Study reference	MSA-26-011	Study date	12/03/2026
Characteristic	PB-1140 pump body spigot OD	Performed by	Priya Nair, Quality Engineer
Specification	12.000 mm (LSL 11.940 / USL 12.060)	Tolerance	0.120 mm
Measurement equipment	Mitutoyo Crysta-Apex S 574, asset AMG-CMM-02	Location	Line 2 Quality Room

Study design

3 operators x 10 parts x 3 trials, randomised measurement order. Analysis performed using the analysis of variance (ANOVA) method. Parts selected to represent the range of product presented at the operation. Raw measurement data held in quality records, ref MSA-26-011.

Results

Source of variation	% of tolerance
Repeatability (equipment variation, EV)	6.9%
Reproducibility (appraiser variation, AV)	5.1%
Total Gauge R&R (%GRR)	8.6%
Number of distinct categories (ndc)	14

%GRR expressed as a percentage of the tolerance (0.120 mm). Repeatability and reproducibility combine to give the total Gauge R&R by the ANOVA method.

Conclusion

The total Gauge R&R of 8.6% of tolerance is below the 10% acceptance criterion. With a number of distinct categories of 14, the measurement system provides adequate resolution for this characteristic. The measurement system is therefore considered acceptable for measurement of the PB-1140 pump body spigot OD on asset AMG-CMM-02.

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